SCD-SAM: Adapting Segment Anything Model for Semantic Change Detection in Remote Sensing Imagery

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***Abstract*— Semantic change detection (SCD) has gradually emerged as a prominent research focus in remote sensing image processing due to its critical role in Earth observation applications. In view of its powerful semantic-driven feature extraction capability, the segment anything model (SAM) has demonstrated its suitability across various visual scenes. However, it suffers from significant performance degradation when confronted with remote sensing images, especially those containing various ground objects that possess significant interclass similarity and substantial intraclass variations. To address the above issues, we propose SCD-SAM, aiming to leverage the potent visual recognition capabilities of SAM for enhanced accuracy and robustness in SCD. Specifically, we introduce a contextual semantic change-aware dual encoder that combines MobileSAM and CNN to extract progressive semantic change features in parallel and inject local features into the MobileSAM encoder through depth feature interaction (DFI) to compensate for the Transformer's limitations in perceiving local semantic details. In addition, in order to utilize the strong visual feature extraction capability of MobileSAM in remote sensing images, we propose a semantic adaptor that aggregates semantic-oriented information about changing objects. To better integrate the extracted contextual semantic information, we devise a progressive feature aggregation dual decoder that aggregates binary change features and semantic change features, respectively, alleviating the semantic gap across different scales. The quantitative and visual results show that SCD-SAM outperforms the state-of-the-art SCD methods on publicly open SCD datasets (e.g., SECOND-CD and Landsat-CD). The code will be available at: https://github.com/yzygit1230/SCD-SAM.**

***Index Terms*— Progressive feature aggregation, segment anything model (SAM), semantic adaptor, semantic change detection (SCD).**

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I. Introduction

CHANGE detection (CD) aims to assess whether various objects within the same region have undergone changes and identify the location of the changed regions by comparing bitemporal images [1], [2], [3]. It finds applications in diverse real-world scenarios, including but not limited to land use renewal, environmental monitoring, cropland management, and damage assessment [4], [5], [6]. Depending on the requirements of the task, CD can be further subdivided into binary CD (BCD) and semantic CD (SCD). BCD provides the automated monitoring and analysis of changing areas, but its information are rough and does not provide specific information about the types of changes occurring. This limitation falls short of meeting the current application requirements for CD in remote sensing images. SCD is an intricate and fine-grained task compared with BCD, which can simultaneously generate changed locations (where) and identify change type (what) in bitemporal images [7]. The results of SCD can provide a richer and nuanced understanding of semantic change, demonstrating the classification of multiple ground objects both pretransformation and posttransformation. Consequently, SCD can provide enhanced decision-making support for practical applications.

SCD entails processing input bitemporal images ($\mathbf{I}_1$ and $\mathbf{I}_2$), and segmenting them semantically to generate bitemporal semantic maps ($\mathbf{S}_1$ and $\mathbf{S}_2$). As shown in Fig. 1, there are four mainstream SCD architectures. In Fig. 1(a), through a

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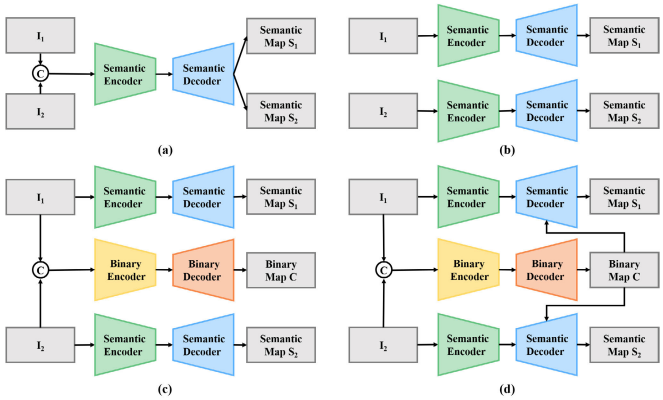


Fig. 1. Overview of four mainstream SCD architectures. (a) Early fusion SCD. (b) Siamese SCD. (c) Separated BCD and SS. (d) Separated BCD and SS connection.

single semantic encoder and semantic decoder, two semantic maps are generated from the concatenation of images $\mathbf{I}_1$ and $\mathbf{I}_2$. Many studies treat SCD as a single semantic segment (SS) task can be considered in this architecture, such as FC-Siam-diff [8]. However, due to the concatenation of bitemporal images in the network input, the exploration of each semantic branch is insufficient, resulting in unsatisfactory performance of SCD. As shown in Fig. 1(b), through the parallel semantic encoder and semantic decoder, two semantic maps are generated from the temporal images $\mathbf{I}_1$ and $\mathbf{I}_2$. For example, PCFNet [9] identified the areas of change and automatically determined change types by leveraging the Siamese architecture. SCDNet [10] improved the ability to detect semantic changes at the boundary of Siamese architecture. Due to the scarcity of changing pixels in SCD, the Siamese network struggles to thoroughly uncover the actual changing region. It also fails to consider binary changes and accurately capture the internal correlations among various ground objects. As shown in Fig. 1(c), some researchers decomposed the SCD task into a BCD subtask and two SS subtasks. This architecture is exemplified by the HRSCD-str.3 [11] and the coarse-to-fine land use CD method [12]. However, the disadvantage of this architecture is the lack of communication between the two SS branches and the BCD branch, which ignores temporal dependencies. As shown in Fig. 1(d), to increase the connection between two SS branches and the BCD branch, some studies involve feeding the binary map into the semantic decoder for feature interaction. For example, HRSCD-str.4 [11] connected the BCD branch to the SS branch by a different skip connection, thus improving the interaction between different branches. ChangMask [13] developed a semantic encoder to learn bitemporal sematic relationship and utilized binary change features to identify change locations. However, the encoders and decoders, which are not designed specifically for BCD and SS, remain inadequate for representing rare binary and semantic changes.

With the continuous advancement of satellite imaging technology, the Earth observation techniques, and the development of deep learning methods, we can now perform CD relatively accurately [14], [15], [16]. Due to the strong sensitivity of the convolutional neural network (CNN) to local change information, it can extract advanced semantic features [17]. Meanwhile, the Transformer can capture the relationships between global pixels in bitemporal images to better model long-range semantic changes [18]. Therefore, the typical CD method involves segmenting semantic change features through the use of Siamese CNNs or Siamese Vision Transformers in an end-to-end manner. Recently, significant advancements have been achieved in foundation models, including GPT-4 [19] and MAE [20], significantly contributing to societal improvement. More recently, vision foundation models (VFMs) have garnered significant attention. Particularly, there has been a surge of interest in the segment anything model (SAM) [21] and its variant MobileSAM [22]. These VFMs are reliable and inclusive, which can generate numerous and comprehensive segmentation masks according to user preferences. The SAM model has been pretrained with the world's largest segmented dataset, which is valuable for SCD. According to different requirements and applications, different targets on bitemporal images, such as vegetation, buildings, forests, and artificial surfaces, may need to be segmented from bitemporal images. This is definitely a significant task that the segmentation "anything" model should not overlook [23].

While significant efforts have been invested, the application of SCD techniques in practical applications continues to be constrained by two primary challenges. On the one hand, semantic change categories are imbalanced, with large differences in scale changes among different categories, making it difficult to extract and represent bitemporal semantic change information. On the other hand, there is significant interclass similarity and substantial intraclass variations in changes of various ground objects, leading to a significant semantic gap. Although SAM is an effective data engine, its robustness to remote sensing images remains limited. As a consequence, SAM encounters difficulty in executing comprehensive target segmentation and identification of remote sensing scenarios. The direct application of SAM to BCD and SCD tasks becomes less generalizable.

Therefore, we exploit the adaption technology for carrying SAM to improve the accuracy in the SCD domain. The main contributions of this work include the following.

1) We propose an SCD-SAM model with a dual-encoder–decoder structure for SCD. In addition, we introduce a dedicated semantic adaptor that aggregates semantically oriented information about change objects in bitemporal images, thereby demonstrating the "segment anything" capabilities of MobileSAM in remote sensing scenes. To the best of the authors' knowledge, this study represents a pioneering effort that introduces and adapts SAM to SCD.
2) We construct a contextual semantic change-aware dual encoder to extract multiscale semantic features. MobileSAM encoder provides a comprehensive understanding of the overall structure and semantic content, extracting rich global semantic change features of various land objects. While the CNN encoder focuses on capturing local detail change information. We also utilize the depth feature interaction (DFI) module to effectively integrate local features into the MobileSAM encoder, allowing each encoder to fully exploit its

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respective strengths, thus complementing each other synergistically.

3) We design a progressive feature aggregation dual decoder to aggregate binary features and bitemporal semantic features separately. The binary decoder generates binary maps to provide prior information for SCD. Siamese semantic decoder aggregates feature at different levels asymptoticly, mitigating the semantic gap between different scales, thereby better understanding the detailed semantic changes. In addition, we devise a contextual semantic discrimination (CSD) module to enhance semantic correlations between Siamese semantic decoders.

II. Related Works

A. Binary Change Detection

Traditional BCD methods often employ thresholding [24], morphological analysis [25], and image algebra [26] for BCD on bitemporal images. Although numerous significant achievements have been made, traditional methods for representing change information are limited and disturbances in the imaging process of remote sensing images (such as viewpoint differences and illumination changes) impede their CD performance [27].

The application of CNN and Transformer to BCD became popular after deep learning gained prominence. CNN is widely used in BCDs due to its fixed convolutional kernel size, which allows for the better extraction of local features of change objects. LGSAANet [28] designed a scale-adaptive attention module to investigate the association between change information and kernel size, as well as skip connections to obtain local pixel correlations. FCCDN [29] proposed a feature constraint network for BCD, which contains a feature pyramid module to extract multiscale features and devises a dense connection module for robust feature fusion. SKTrans [30] introduced a tone shift mechanism to reduce color and structural similarity differences between buildings, and proposed a knowledge integration module to alleviate semantic confusion between background and building objects. CNN excels at extracting advanced semantic features that reveal interesting changes, but its focus on local modeling makes it less robust to wide-range changes. While Transformer is able to capture long-range spatiotemporal information from bitemporal images, which is important for determining the position and shape of changing objects. ACAHNet [31] introduced an asymmetric cross-attention module to preserve the crucial features by incorporating semantic maps, and proposed a mid-term fusion strategy for feature fusion. MDAFormer [32] utilized a multilevel Transformer to acquire bitemporal change representations and employed a difference aggregation module to generate discriminative feature maps. DiFormer [33] proposed a token difference evaluation module to generate the discrepancy between the change object and the background and developed a multiscale semantic perception module to enable difference modeling. However, Transformer-based methods largely neglect the static receptive fields of individual token features, resulting in inadequacy regarding the edges of the change of interest.

B. Semantic Change Detection

In contrast to BCD, which solely discerns only between changed and unchanged areas, SCD provides category transition information, making it more useful in remote sensing applications as it detects not only the "where" but also the "what" of changes. ReCNN [34] used CNN to achieve combined spectral–spatial feature extraction and employed recurrent neural network (RNN) to learn multiclass changes adaptively. HRSCD [11] compared and analyzed four strategies (str.1–str.4) of SCD systematically. Among them, HRSCD-str.4 achieved the best results, indicating that temporal correlation can reduce the occurrence of missing detection of semantic changes. ASNet [35] mitigated semantic ambiguity by considering the differences in the distribution of various land features in bitemporal images and created a large-scale benchmark SCD dataset.

Since the significant interclass similarity and substantial intraclass variations in various semantic changes, pseudochanges exert a considerable negative impact on SCD. Consequently, some efforts have been made to alleviate the influence of pseudochanges by incorporating multiscale feature fusion into the network. HPCFNet [36] proposed a multipart feature learning strategy to detect multiclass changes by using Siamese CNN to fuse and stratify multiscale features. KGLPD [37] proposed an adaptive gradient sensing mechanism to provide prior knowledge of block distribution and designed a cognitive land pattern model to detect relationships between different ground objects. Y-Net [38] employed a dual-stream network to extract bitemporal change features and a full-scale skip connection algorithm to reconstruct the multilevel change maps. Although the above methods improve the performance of SCD to some extent, the learning of temporal dependencies remains partial. It is necessary to bridge the gaps between bitemporal semantic changes and to consider significant binary changes, which is one of the goals of this article.

C. Vision Foundation Models

The advent of VFMs has ushered in a transformative era in deep learning. In addition, VFMs exhibit impressive zero-shot generalization capabilities across diverse scenarios. Numerous researchers have explored the use of pretrained vision models on large-scale datasets to achieve generic identification capabilities applicable to subsequent applications. For instance, MAE [20] first proposed a Transformer-based visual masking training strategy, which achieved outstanding self-supervision effects by extensively masking a high proportion of input images. CLIP [39] trained with a large number of image–text pairs, demonstrates the effective enhancement of similarity between images and text through metric learning. SAM-CD [23] utilized the visual encoder of VFM to extract bitemporal change features and introduced a convolutional adapter to aggregate feature information tailored for BCD. RingMo-SAM [40] collected millions of segmentation instances from multiple multimodal datasets to train VFM and developed a category-decoupling mask decoder for multimodal remote sensing image segmentation. SkySense [41] pretrained on a multimodal remote sensing image dataset containing

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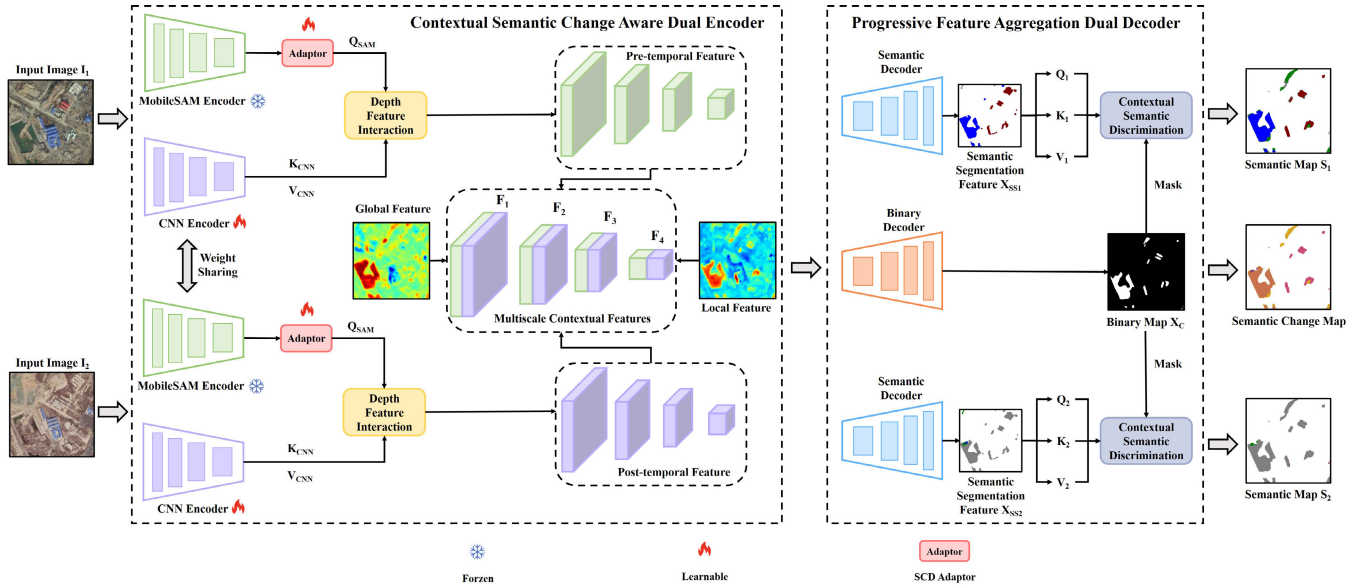


Fig. 2. Overview of the proposed SCD-SAM.

21.5 million time series using multiscale contrastive learning to learn representations of different modalities and spatial granularities.

A burgeoning trend in VFMs is the investigation of establishing foundational models that can generalize across a myriad of common visual tasks. SAM [21] is a VFM that has been trained on potentially millions of natural images, allowing for zero-shot generalization to objects and images not encountered during training. To further improve the computational efficiency of SAM, MobileSAM [22] was able to maintain SAM performance while reducing the model size by more than 60 times. Although SAM has demonstrated outstanding results on natural images, it has encountered some challenges when attempting to segment remote sensing images, particularly for ground objects with varying scales, complex shapes, and blurred boundaries [42]. Thus, in this article, we explore the applicability of SAM to SCD by utilizing adaptation.

III. Methodology

A. Overview of SCD-SAM

On the basis of the above considerations, we further strengthen the connection between each branch based on the concept of separate BCD and SS and design a dual-encoder–decoder architecture. Specifically, SCD-SAM receives bitemporal images as input and includes three primary components, as shown in Fig. 2. The contextual semantic change-aware dual encoder extracts bitemporal semantic features, while the progressive feature aggregation dual decoder generates both binary change features and two semantic change features simultaneously, and a CSD module further discriminates cross-temporal SS features based on bitemporal semantic change features. Finally, SCD-SAM outputs two semantic maps and a semantic change map.

B. Architecture of SCD-SAM

1) Contextual Semantic Change-Aware Dual Encoder: There is a significant imbalance between the semantic categories in the remote sensing images, with a high degree of similarity between categories and a substantial degree of diversity within categories. To obliterate this obstacle, we devise a contextual semantic change-aware dual encoder to extract latent information present in remote sensing images and characterize changes in various types of land objects. As shown in Fig. 3, contextual semantic change-aware dual encoder inherits the architecture from MobileSAM, undergoes modifications to the image encoder and introduces a CNN branch to address challenges associated with insufficient local features, making it more suitable for SCD. Our semantic adaptor includes reducing the input size, modifying overlapping patch embedding, introducing multiscale semantic features, and incorporating a DFI module.

As VFMs face challenges in effectively discriminating ground objects in bitemporal remote sensing images. To better suit the applied SCD task in this study, we resize the input images for MobileSAM from 1024 × 1024 pixels to 512 × 512 pixels. The overlapping patch embedding shares the same parameters as the original MobileSAM, but its patch stride is half of the original stride, effectively capturing semantic context at the patch boundaries. Since MobileSAM is constructed with Transformer, which dynamically focuses on information in the input sequence through attention weights, its feature maps are 3-D ($B \times N \times C$). We have developed a simple yet effective semantic adaptor to improve the preservation of high-order semantic features extracted from MobileSAM. It converts the original 3-D feature maps into a 4-D feature map ($B \times C \times H \times W$). The addition of multiscale information also enhances the ability of the network to handle scale variations by providing a more comprehensive context for semantic changes. Therefore, we modify MobileSAM into four stages to capture multiscale semantic change features, with channel numbers of 96, 192, 384, and 768. This modification is intended to improve the semantic perception of MobileSAM for different types of land cover change.

The local features extracted by the CNN have been shown to be robust to boundary blur and shadow interference

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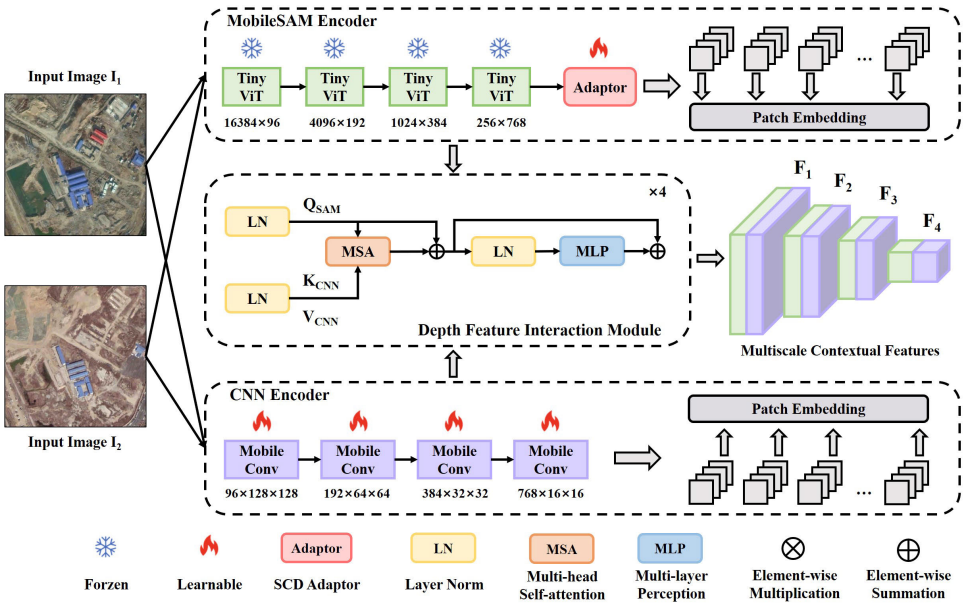


Fig. 3. Overview of the contextual semantic change-aware dual encoder.

between bitemporal images. Therefore, our CNN encoder stacks four mobileConv modules, each comprising sequentially connected batch normalization functions, convolutional blocks, and Gaussian error linear unit (GELU) activation functions. Specifically, the input is initially passed through a 1 × 1 convolutional block to expand its channel count to four times the original size. Then, a 3 × 3 depthwise convolution captures local correlations. Finally, a 1 × 1 convolution projects the features to the original channel size, facilitating residual connections. It is noteworthy that the multiscale feature maps generated by the four MobileConv modules maintain consistency with the channels and spatial resolution of the outputs of the MobileSAM encoder.

To better calculate and project the CNN encoder responses for multiscale features in the MobileSAM encoder, we design a DFI module to serve as a bridge between the MobileSAM encoder and the CNN encoder, enhancing the integration of missing local features with MobileSAM. Specifically, we initially reshape the depth features from the MobileSAM encoder and CNN encoder into flattened patches. Subsequently, we project them into token vectors, denoted by $\mathbf{T}_{\mathrm{SAM}}$ and $\mathbf{T}_{\mathrm{CNN}}$, respectively. In each attention layer, we compute global query $\mathbf{Q}_{\mathrm{SAM}}$ from the input $\mathbf{T}_{\mathrm{SAM}}$, while compute local key $\mathbf{K}_{\mathrm{CNN}}$ and local value $\mathbf{V}_{\mathrm{CNN}}$ from the input $\mathbf{T}_{\mathrm{CNN}}$ as follows:

$$\begin{aligned}\mathbf{Q}_{\mathrm{SAM}} &= \mathbf{T}_{\mathrm{SAM}}\mathbf{W}_{\mathrm{q}}\\ \mathbf{K}_{\mathrm{CNN}} &= \mathbf{T}_{\mathrm{CNN}}\mathbf{W}_{\mathrm{k}}\\ \mathbf{V}_{\mathrm{CNN}} &= \mathbf{T}_{\mathrm{CNN}}\mathbf{W}_{\mathrm{v}}\end{aligned} \tag{1}$$

where $\mathbf{W}_{\mathrm{q}}$, $\mathbf{W}_{\mathrm{k}}$, and $\mathbf{W}_{\mathrm{v}}$ are different linear projection weights. Subsequently, the enhanced global token $\tilde{T}_{\mathrm{SAM}}$ is calculated as follows:

$$\tilde{T}_{\mathrm{SAM}} = \mathrm{softmax}\left(\frac{\mathbf{Q}_{\mathrm{SAM}}\mathbf{K}_{\mathrm{CNN}}^{\mathbf{T}}}{\sqrt{d}}\right)\mathbf{V}_{\mathrm{CNN}} + \mathbf{T}_{\mathrm{SAM}} \tag{2}$$

where softmax() denotes the softmax function and d is the channel dimension of $\mathbf{Q}_{\mathrm{SAM}}$ and $\mathbf{K}_{\mathrm{CNN}}$. Then, the DFI module improves the modeling capability of complex relationships between the enhanced global tokens as follows:

$$\hat{T}_{\mathrm{SAM}} = \mathrm{MLP}\big(\mathrm{LN}\big(\tilde{T}_{\mathrm{SAM}}\big)\big) + \tilde{T}_{\mathrm{SAM}} \tag{3}$$

where MLP() denotes the multilayer perception function and LN() denotes the LayerNorm function. The above operations will be repeated four times. As a result, the local tokens and global tokens will have sufficient interaction to allow the CNN encoder to inject local features into the MobileSAM encoder.

2) Progressive Feature Aggregation Dual Decoder: Due to different scales and morphologies of different land cover objects in remote sensing images, which makes SCD challenging. The direct fusion of the extracted multiscale features does not take into account continuous contextual semantic information, which results in a reduced perception of semantic change edges. Therefore, as shown in Fig. 4, our proposed progressive feature aggregation dual decoder employs a Siamese semantic decoder to augment the consideration of continuous contextual semantic information, and simultaneously utilizes an independent binary decoder to generate an additional binary change map, offering further insights into change discrimination.

As input, progressive feature aggregation dual decoder takes the four multiscale semantic change features obtained by contextual semantic change-aware dual encoder, with spatial sizes of 128 × 128, 64 × 64, 32 × 32, and 16 × 16, respectively, which can represent semantic changes at different scales and types of land cover.

Since BCD only needs to determine whether changes have occurred in bitemporal images, and the deep feature maps contain more rich features related to changes in multiple land types. Therefore, we progressively upsample the deepest feature map, $\mathbf{F}_4$, layer by layer and aggregate it with its upper layer feature maps at each stage, ultimately aggregating to the shallowest feature map, $\mathbf{F}_1$. Through the aggregation of multiscale features in the binary decoder, the deep feature

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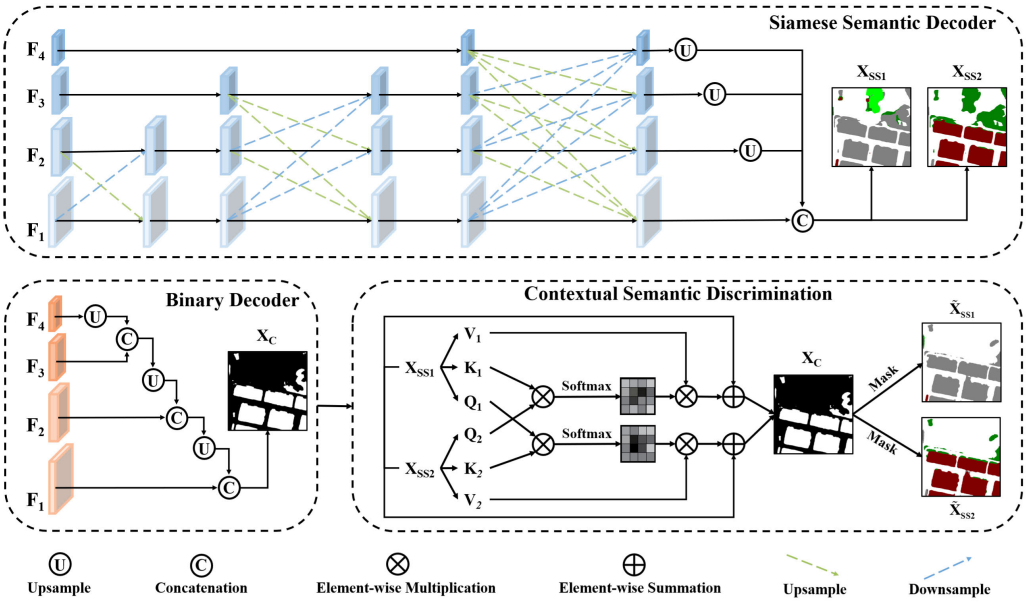


Fig. 4. Overview of the progressive feature aggregation dual decoder.

maps rich in semantic information generate the binary change map $\mathbf{X}_c$, which can be used as prior information to guide SCD.

For SCD, a more detailed discrimination of semantic change types and positions at the pixel level is required. Thus, it is necessary to fully utilize the extracted multiscale feature information. Directly aggregating the features of the four scales may result in the loss of some fine features, resulting in a reduction in the robustness of the network to noise. Therefore, our semantic decoder adopts a progressive strategy to make the semantic information of features at different levels approach each other during the asymptotic fusion process. Specifically, the semantic decoder first interacts with the shallowest features ($\mathbf{F}_1$ and $\mathbf{F}_2$), then aggregates the deeper features ($\mathbf{F}_3$), and finally fuses the deepest features ($\mathbf{F}_4$). Finally, the thoroughly interacted four feature maps are concatenated to output the semantic segmentation features. Since the semantic decoder has a Siamese structure, it can simultaneously output the pretemporal semantic segmentation feature $\mathbf{X}_{SS1}$ and the posttemporal semantic segmentation feature $\mathbf{X}_{SS2}$. In this way, we are capable of reducing the substantial semantic disparity that exists between levels that are not directly contiguous.

Although the semantic decoder possesses a Siamese structure, there still exists a temporal asymmetry issue. In SCD, considering semantic correlations between bitemporal images is crucial. To this end, we devise a CSD module to perform cross-temporal discrimination on the fused bitemporal SS feature maps. The CSD module first linearly maps the input SS maps $\mathbf{X}_{SS1}$ and $\mathbf{X}_{SS2}$ to obtain $\mathbf{Q}_1$, $\mathbf{K}_1$, and $\mathbf{V}_1$ for $\mathbf{X}_{SS1}$, and $\mathbf{Q}_2$, $\mathbf{K}_2$, and $\mathbf{V}_2$ for $\mathbf{X}_{SS2}$. Subsequently, it performs a matrix multiplication between $\mathbf{Q}_1$ and $\mathbf{K}_2$, generates an attention matrix through the softmax activation function. Then, this attention matrix is multiplied with $\mathbf{V}_2$ and added to the input SS map $\mathbf{X}_{SS1}$ to obtain the enhanced discriminated semantic segmentation map $\tilde{X}_{SS1}$. The corresponding operations for the input $\mathbf{X}_{SS2}$ are symmetrically performed, which is calculated as follows:

$$\begin{aligned}\tilde{X}_{SS1} &= \mathbf{X}_{SS1} + \text{softmax}(\mathbf{Q}_1 \times \mathbf{K}_2) \times \mathbf{V}_2 \\ \tilde{X}_{SS2} &= \mathbf{X}_{SS2} + \text{softmax}(\mathbf{Q}_2 \times \mathbf{K}_1) \times \mathbf{V}_1 \end{aligned} \tag{4}$$

where $\mathbf{X}_{SS1}$ and $\mathbf{X}_{SS2}$ are the input SS maps, $\tilde{X}_{SS1}$ and $\tilde{X}_{SS2}$ are the enhanced discrimination SS maps. It is intuitive that the CSD module motivates and supports a temporal image to contemplate the semantic features of opposite temporal maps while preserving its original semantic characteristics, efficiently bridging bitemporal semantic information flow and exploiting cross-temporal contextual features.

C. Objective Function of SCD-SAM

We utilize three loss functions to fully supervise the learning of the proposed SCD-SAM, including SS loss L_{SS}, binary change loss L_{BCD}, and semantic change loss L_{SCD}.

SS loss L_{SS} calculates the multiclass loss between the SS result $\mathbf{S}_1$ and the SS label $\mathbf{G}_{SS_1}$, as well as between the SS result $\mathbf{S}_2$ and the SS label $\mathbf{G}_{SS_2}$. L_{SS} for each pixel is calculated as follows:

$$L_{SS} = -\frac{1}{N}\sum_{i=1}^{N} \mathbf{G}_{SS_i} \log\left(\mathbf{P}_{SS_i}\right) \tag{5}$$

where N is the number of semantic categories and $\mathbf{P}_{SS_i}$ and $\mathbf{G}_{SS_i}$ are the predicted and label values of the ith semantic category, respectively.

Binary change loss L_{BCD} calculates the binary loss between BCD result C and BCD label. L_{BCD} for each pixel is calculated as follows:

$$L_{BCD} = \mathbf{G}_{BCD} \log(\mathbf{P}_{BCD}) + (\mathbf{1} - \mathbf{G}_{BCD}) \log(\mathbf{1} - \mathbf{P}_{BCD}) \tag{6}$$

where $\mathbf{P}_{BCD}$ and $\mathbf{G}_{BCD}$ represent the predicted and label value of BCD, respectively.

To align the bitemporal semantic change and binary change information in the SCD task, thereby enhancing the integration

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of SS and BCD subtasks. We use semantic change loss L_{SCD} to calculate the cosine loss between SS prediction results $\mathbf{S}_1$ and $\mathbf{S}_2$ and corresponding BCD label $\mathbf{G}_{\mathrm{BCD}}$

$$L_{\mathrm{SCD}} = \begin{cases} 1 - \cos(\mathbf{M}_1, \mathbf{M}_2), & \mathbf{P}_{\mathrm{BCD}} = 1 \\ \cos(\mathbf{M}_1, \mathbf{M}_2), & \mathbf{P}_{\mathrm{BCD}} = 0 \end{cases} \tag{7}$$

where $\mathbf{M}_1$ and $\mathbf{M}_2$, respectively, represent the pixel-level semantic feature maps on the semantic segmentation results $\mathbf{S}_1$ and $\mathbf{S}_2$.

IV. Experiments

A. Datasets

1) SECOND-CD: The SECOND-CD [35] dataset is an extensive and meticulously annotated dataset for SCD, which comprises 2968 high-resolution images obtained from various sensors. These images capture diverse urban landscapes, including Hangzhou, Chengdu, and Shanghai in China. Each image has a size of 512 × 512 pixels, with resolutions ranging from 0.3 to 5 m. The dataset incorporates seven semantic classes, encompassing an unchanged class and six land cover semantic classes (water, ground, low vegetation, tree, building, and playground). We further divided the dataset into a training set with 2375 images and a test set with 593 images.

2) Landsat-CD: The Landsat-CD [43] dataset is constructed using Landsat imagery spanning the years 1990–2020 and is located in Tumushuke, Xinjiang, China. This dataset is segmented into an unchanged class and four categories representing semantic changes (farmland, desert, building, and water). Comprising a total of 2385 original image pairs, each possessing size of 416 × 416 pixels and a resolution of 30 m. We further divided the dataset into a training set with 1908 images and a test set with 477 images.

B. Experimental Settings

We conduct all experiments using the PyTorch framework on the NVIDIA A100 GPU with memory of 80 GB. We train all models using the same experimental parameters, including a batch size of 8. All methods are performed for 100 epochs of training. In addition, we employ AdamW to update the gradient, with an initial learning rate of 0.00035.

We use four evaluation metrics to evaluate SCD methods, including overall accuracy (OA), mean intersection over union (mIOU), separated kappa (SeK), and variants of the F1 metric in BCD ($\mathrm{F}_{\mathrm{SCD}}$) [44]. Assuming a confusion matrix $R = \{r_{x,y}\}$, where $r_{x,y}$ represents the number of pixels classified in class x, and y represents the corresponding label index ($x, y \in \{0, 1, \ldots, n\}$) (note that 0 represents unchanged). OA is calculated as follows:

$$\mathrm{OA} = \sum_{x=0}^{n} r_{x,x} \Big/ \sum_{x=0}^{n}\sum_{y=0}^{n} r_{x,y}\,. \tag{8}$$

However, the equivalence of each pixel in the OA calculation will yield an unjustifiable score, especially when dealing with a predominant unchanged pixel. Therefore, we combine unchanged IOU_0 and changed IOU_1 to calculate the mIOU, thereby enhancing the discrimination between unchanged and changed pixels

$$\mathrm{IOU}_0 = r_{0,0} \Big/ \left(\sum_{x=0}^{n} r_{x,0} + \sum_{y=0}^{n} r_{0,y} - r_{0,0} \right) \tag{9}$$

$$\mathrm{IOU}_1 = \sum_{x=1}^{n}\sum_{y=1}^{n} r_{x,y} \Big/ \left(\sum_{x=0}^{n}\sum_{y=0}^{n} r_{x,y} - r_{0,0} \right) \tag{10}$$

$$\mathrm{mIOU} = (\mathrm{IOU}_0 + \mathrm{IOU}_1)/2 \tag{11}$$

where n is the total number of classifications including the unchanged class. Due to the significant proportion of unchanged pixels, the impact on the result cannot be ignored. Therefore, we adopt SeK to mitigate the label imbalance problem. Given the matrix $R' = \{r'_{x,y}\}$, where $r_{x,y} = r'_{x,y}$, except for $r'_{x,y} = 0$. SeK is calculated as follows:

$$\rho = \sum_{x=0}^{n} r'_{x,x} \Big/ \sum_{x=0}^{n}\sum_{y=0}^{n} r'_{x,y} \tag{12}$$

$$\eta = \sum_{x=1}^{n} \left(\sum_{y=1}^{n} r'_{x,y} \cdot \sum_{y=1}^{n} r'_{y,x} \right) \Big/ \left(\sum_{x=1}^{n}\sum_{y=1}^{n} r'_{x,y} \right)^2 \tag{13}$$

$$\mathrm{SeK} = e^{\mathrm{IOU}_1 - 1} \cdot (\rho - \eta)/(1 - \eta). \tag{14}$$

Inspired by the $F1$ metric in BCD, to provide a more intuitive evaluation of the classification of different ground objects in SCD, we introduce the F_{SCD} metric, which is calculated as follows:

$$P_{\mathrm{SCD}} = \sum_{x=1}^{n} r_{x,x} \Big/ \sum_{x=1}^{n}\sum_{y=0}^{n} r_{x,y} \tag{15}$$

$$R_{\mathrm{SCD}} = \sum_{x=1}^{n} r_{x,x} \Big/ \sum_{x=0}^{n}\sum_{y=1}^{n} r_{x,y} \tag{16}$$

$$\mathrm{F}_{\mathrm{SCD}} = \frac{2 \times P_{\mathrm{SCD}} \times R_{\mathrm{SCD}}}{P_{\mathrm{SCD}} + R_{\mathrm{SCD}}} \tag{17}$$

where P_{SCD} and R_{SCD} represent the variants of precision and recall, respectively.

C. Integrated Visual Analysis

Fig. 5 illustrates the bitemporal SS, BCD, and SCD results of SCD-SAM on the SECOND-CD. Since SS labels between bitemporal images maintain background consistency, as they include the same nochange class. As a result, semantic properties of pretemporal objects can be found in corresponding properties of posttemporal objects, and vice versa. Therefore, we generate 31 classes of SCD labels corresponding to the existing seven classes of SS labels in the SECOND-CD. Overall, our SS results maintain a high level of consistency with the SS labels, this is because SCD-SAM looks for commonalities between similar change classes (i.e., low vegetation and tree), while learning from differences between different change classes (i.e., ground and building). From the three predicted maps and attention maps of BCD, it can also be observed that the semantically driven VFM exhibits robust capability in representing change features, effectively focusing

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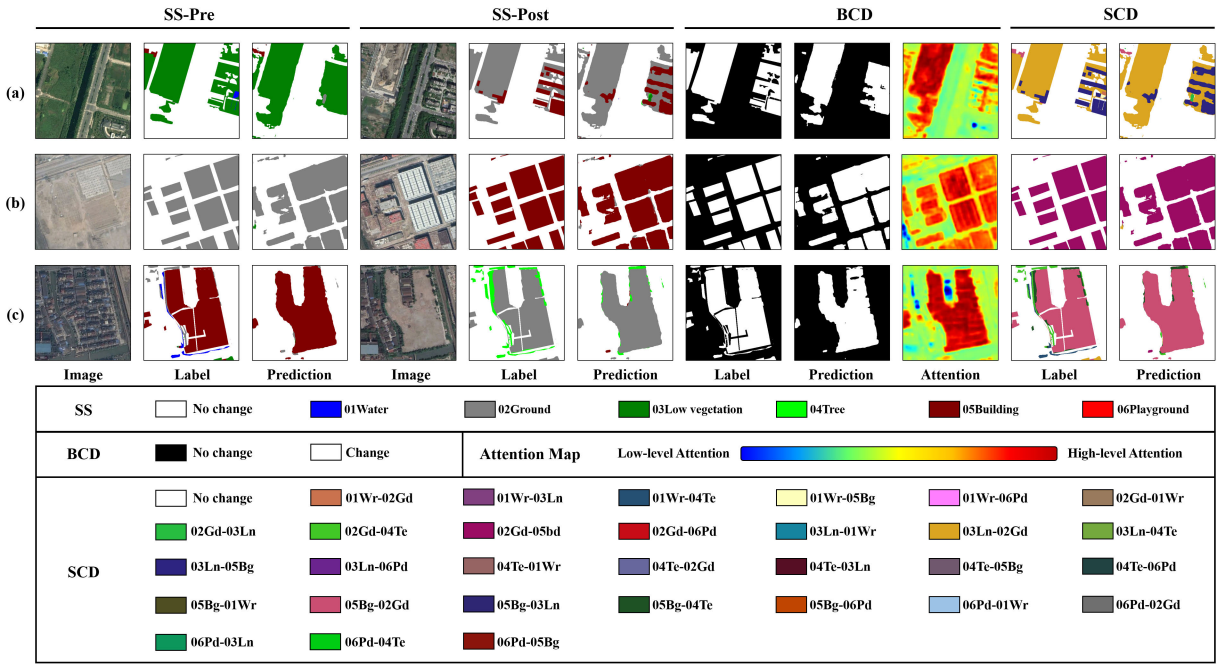


Fig. 5. Bitemporal SS, BCD, and SCD results in (a)–(c) three typical scenes from the SECOND-CD.

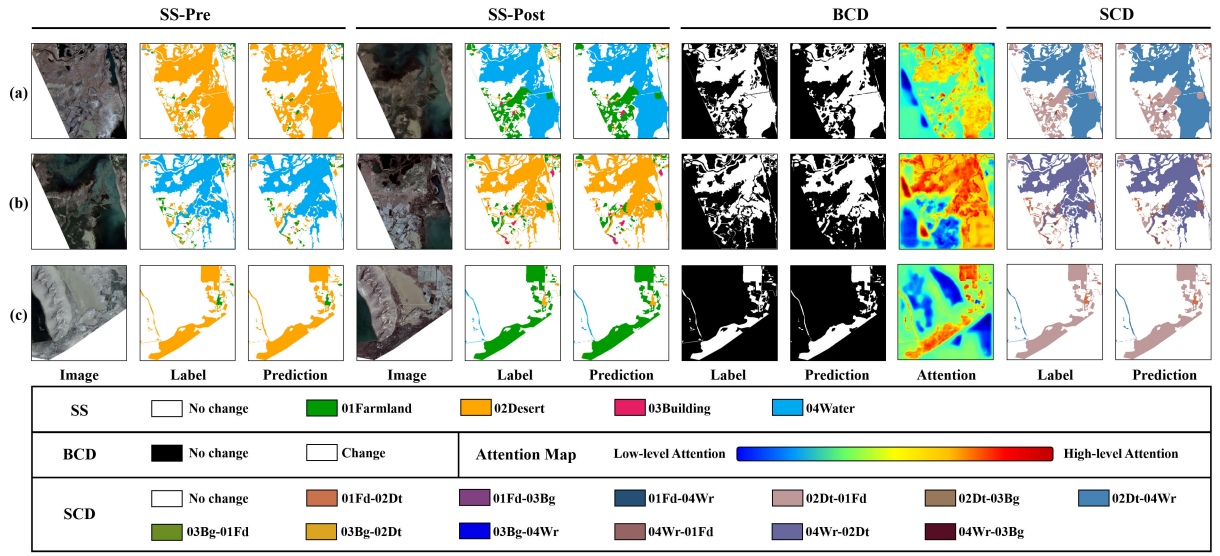


Fig. 6. Bitemporal SS, BCD, and SCD results in (a)–(c) three typical scenes from the Landsat-CD.

on regions with actual changes. In addition, due to the dual-decoder structure, despite the presence of up to 31 classes of changes in SCD, SCD-SAM is still capable of detecting subtle differences between different types of changes, ensuring the accuracy of SCD results.

As shown in Fig. 6, SCD-SAM can produce more accurate and sharper SS, BCD, and SCD results than SECOND-CD in the Landsat-CD, especially for the Farmland edge. While the majority of SS categories are accurately segmented, there are still some areas of overestimation, such as the Desert in SS-pre category and the water category in SS-post, as shown in Fig. 6(a). We can intuitively find that SCD-SAM brings incremental performance improvements to BCD results. It boosts the detection of slight changes, such as building changes across bitemporal images in Fig. 6(a) and (b). From the three attention maps, it is evident that SCD-SAM exhibits a robust ability to discriminate semantic objects of different scales and categories amidst complex backgrounds. By transforming the original five SS labels, we generate a total of 13 SCD labels for SCD analysis. Fig. 6(a) and (b) demonstrates that SCD-SAM exhibits inspiring semantic change perception for large areas of desert–water and the corresponding water–desert semantic changes. In addition, due to the inclusion of the CNN branch in the dual-encoder structure, the detailed change representation performance is enhanced, enabling the SCD of spindly objects, as shown in Fig. 6(c).

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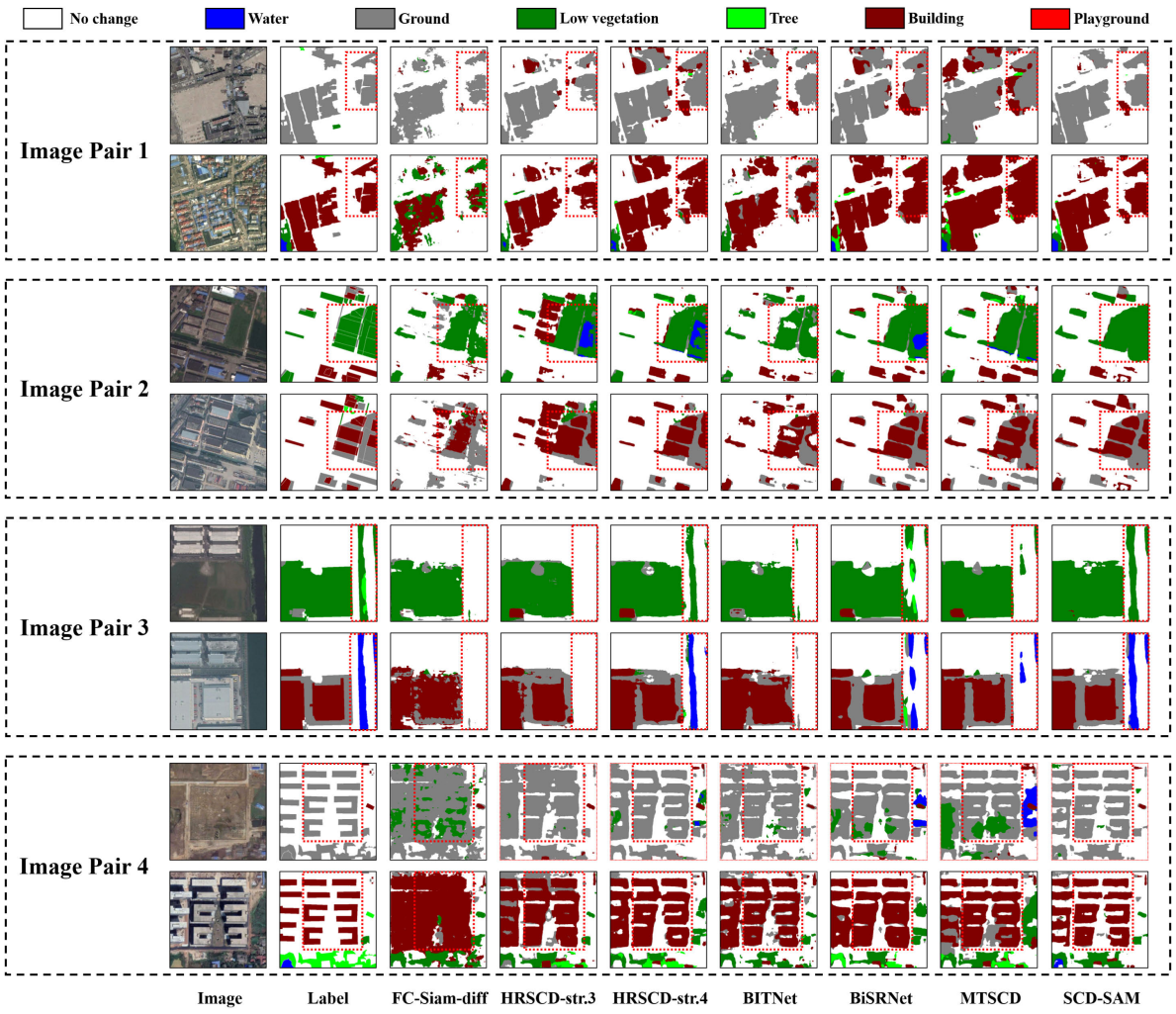


Fig. 7. Visual detection results of various comparative methods from four typical scenes from the SECOND-CD.

TABLE I

QUANTITATIVE COMPARISON RESULTS ON THE SECOND-CD. THE **RED** AND **BLUE** HIGHLIGHTS INDICATE THE OPTIMAL AND SUBOPTIMAL VALUES, RESPECTIVELY

Method	OA(%)	mIOU(%)	SeK(%)	F_{SCD}(%)
FC-Siam-diff [8]	81.54	61.93	5.72	44.65
HRSCD-str.3 [11]	84.34	66.78	12.42	52.51
HRSCD-str.4 [11]	86.65	71.34	18.91	58.33
BITNet [45]	84.55	67.12	11.39	50.85
BiSRNet [44]	87.02	72.03	21.28	61.02
MTSCD [7]	**87.02**	**72.13**	**21.70**	**61.57**
SCD-SAM	**90.16**	**77.75**	**32.44**	**70.51**

D. Comparative Results

1) Comparison Methods: To perform a holistic assessment of the proposed SCD-SAM, we perform quantitative and visual analysis with several advanced SCD methods, including FC-Siam-diff [8], HRSCD-str.3 [11], HRSCD-str.4 [11], BITNet [45], BiSRNet [44], and MTSCD [7].

2) Quantitative Analysis: Table I indicates that SCD-SAM achieves the best quantitative results on the SECOND-CD. Notably, FC-Siam-diff exhibits the worst performance, indicating that the direct fusion approach employed by the Siamese network for SCD is significantly affected by class imbalance. Comparatively, by decomposing SCD into two distinct subtasks of BCD and SS, HRSCD-str.4, BiSRNet, and SCD-SAM mitigate class imbalance issues and obtain higher mIOU and SeK. Although BITNet is significantly outperforming HRSCD-str.3 in the mIOU metric, its SeK value is 1.03% lower, indicating that the increase in SeK is not strongly correlated with BCD performance. This is based on mIOU measuring the results of no-change and change pixel detection, while SeK measures the accuracy of all changed semantic categories. In addition, the remarkable achievement lies in the fact that SCD-SAM shows improvements in OA/mIOU/SeK/F_{SCD} of 3.14%/5.62%/10.74%/8.94% compared with the current best-performing network (MTSCD) on the SECOND-CD. This implies that employing different BCD and SCD decoders could enhance the multiscale fusion of potential semantic changes within the network.

Table II provides the four quantitative evaluation results obtained from the Landsat-CD. Since the number of semantic change labels has increased, various methods have demonstrated significant improvements in the quantitative metrics of

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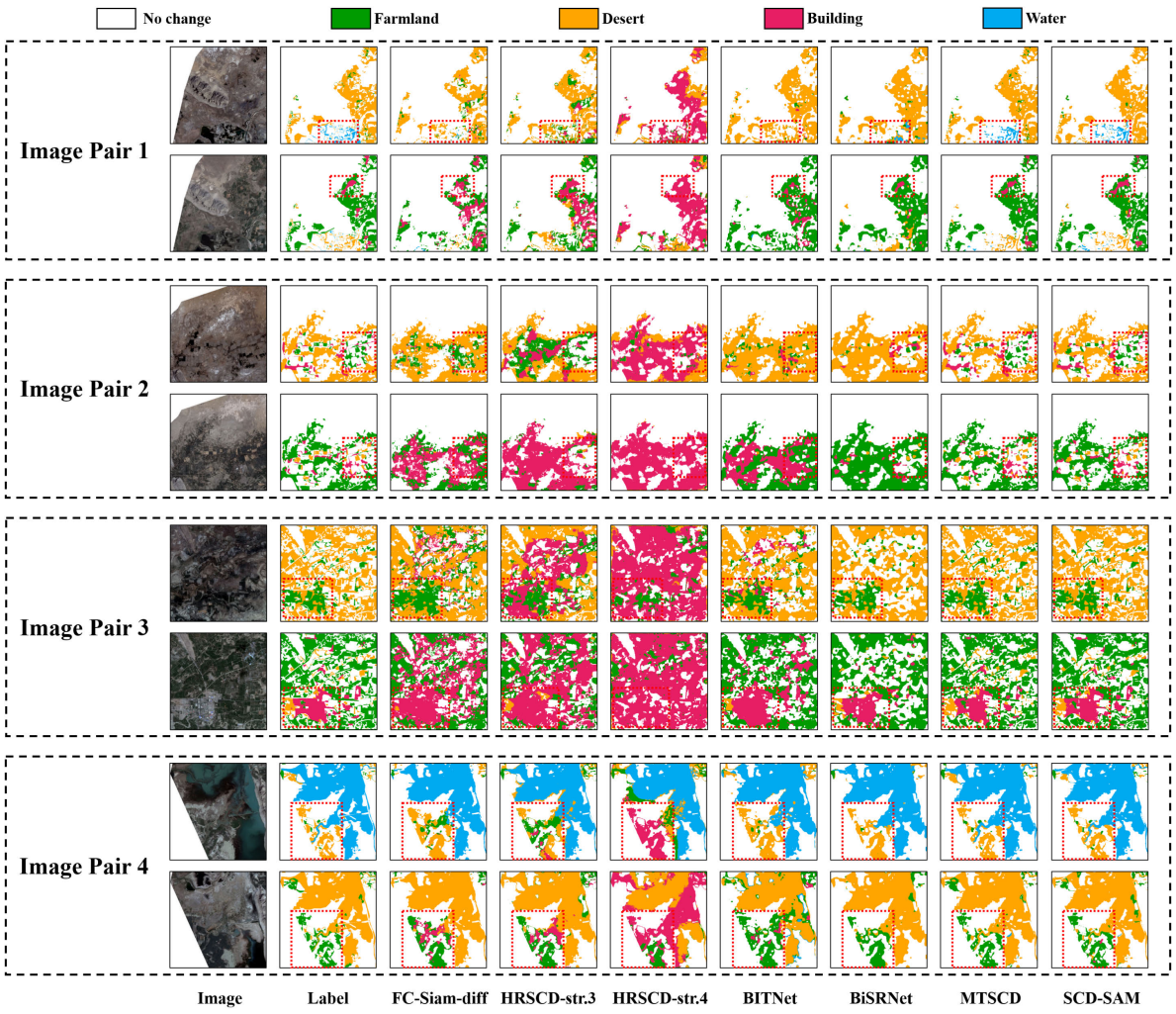


Fig. 8. Visual detection results of various comparative methods from four typical scenes from the Landsat-CD.

TABLE II

QUANTITATIVE COMPARISON RESULTS ON THE LANDSAT-CD. THE **RED** AND **BLUE** HIGHLIGHTS INDICATE THE OPTIMAL AND SUBOPTIMAL VALUES, RESPECTIVELY

Method	OA(%)	mIOU(%)	SeK(%)	F_{SCD}(%)
FC-Siam-diff [8]	89.38	73.89	24.47	69.07
HRSCD-str.3 [11]	90.05	77.86	30.04	71.12
HRSCD-str.4 [11]	88.41	77.51	21.78	62.71
BITNet [45]	87.89	71.92	20.26	64.93
BiSRNet [44]	91.24	78.60	33.73	75.11
MTSCD [7]	**92.76**	**81.43**	**40.79**	**79.73**
SCD-SAM	**93.30**	**83.04**	**44.05**	**81.16**

SCD compared with the SECOND-CD. Among the methods compared, BCD-oriented approaches achieve unsatisfactory results. For example, the SeK values of FC-Siam-diff and BITNet are only 24.47% and 20.26%, respectively, which are much inferior to other SCD-oriented methods. HRSCD-str.4 contains skip connections between the Siamese SS branch and BCD branch, mitigating the drawbacks of the standard SCD method, with an mIOU value of 77.51%. BiSRNet facilitates the interaction between the bitemporal SS branches, which achieve suboptimal quantitative results, with SeK and FSCD values of 33.73% and 75.11%, respectively. Furthermore, our proposed SCD-SAM improves the depth interaction of local and global features in bitemporal SS branches, significantly outperforming the compared methods in all metrics—achieving 93.30%, 83.04%, 44.05%, and 81.16% in OA, mIOU, SeK, and F_{SCD}, respectively. This confirms that SAM can be effectively applied to SCD in remote sensing images through sufficient and effective adaptation.

3) Visual Analysis: Fig. 7 presents the visual comparisons among various methods in four typical scenes using the SECOND-CD. SCD-SAM maintains more accurate boundaries between different semantic classes compared with the comparison methods in four image pairs, which illustrates the preservation of more valuable multiscale features to enhance bitemporal SS performance. It is obvious that FC-Siam-diff lacks sufficient detection capability for semantic changes, leading to numerous regions with both false detections and missed detections, particularly noticeable in image pair 4. While the Transformer can globally model various semantic changes in images, it lacks the ability to distinguish fine details of objects. This limitation leads to suboptimal SCD results for BITNet. In image pair 3, only HRSCD-str.4, BiSRNet,

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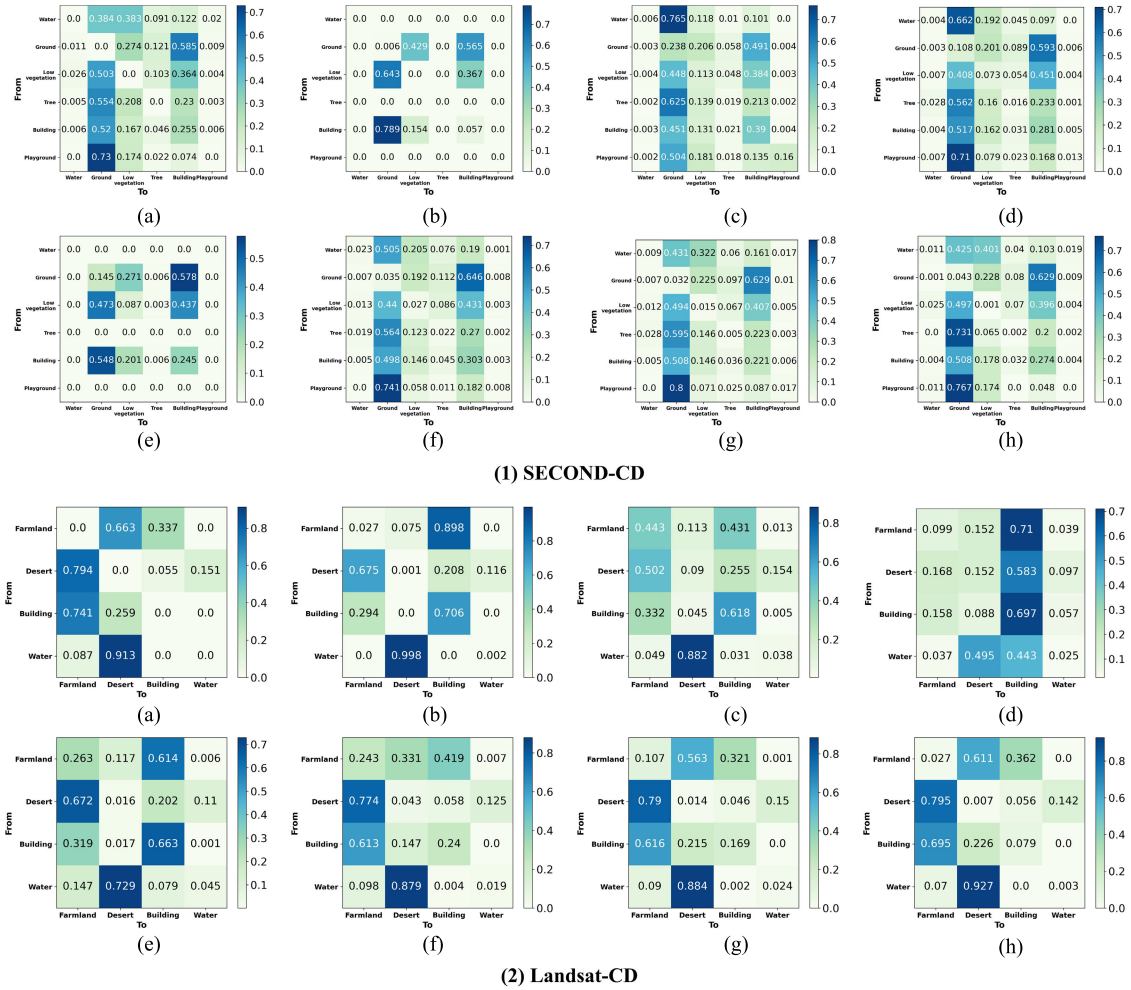


Fig. 9. "From–to" matrices of various comparative methods from: 1) SECOND-CD and 2) Landsat-CD. (a) Ground truth. (b) FC-Siam-diff. (c) HRSCD-str.3. (d) HRSCD.str.4. (e) BITNet. (f) BiSRNet. (g) MTSCD. (h) SCD-SAM.

MTSCD, and our proposed method are capable of detecting semantic changes in two types of land cover, vegetation, and water. This further demonstrates the effectiveness of the separate BCD and SS strategies. Meanwhile, SCD-SAM successfully captures all these semantic change types, further confirming its nuanced ability to discriminate semantic land changes in remote sensing scenes.

Fig. 8 displays the visual comparisons across different methods in four representative scenes utilizing the Landsat-CD. It is evident that SCD-SAM adeptly identifies changed regions, distinguishing between various semantic change classes with well-defined edges and maintaining high internal compactness. It can be observed in image pair 1, that due to the powerful semantic representation capability of SAM, only SCD-SAM can detect semantic changes in the water region in the pretemporal image and desert in the posttemporal image. In addition, from image pairs 2 and 3, BiSRNet tends to overestimate the size of desert and farmland objects, predicting them with a significantly larger area. Based on the results of four image pairs, HRSCD-str.4 contains a significant number of pink pixels, which represent erroneously detected areas. This demonstrates a clear limitation to its perception capability in desert regions. In addition, since the DFI module can incorporate local details extracted from the CNN encoder, SCD-SAM demonstrates strengths in discriminating small land objects in image pairs 2 and 4.

4) "From–To" Semantic Change Analysis: Fig. 9 illustrates the "From–To" matrices generated by different methods, providing a graphical representation of the "From–To" semantic change types in the SECOND-CD and Landsat-CD. In the SECOND-CD, the primary semantic changes entail transitions "Playground–Ground" and "Ground–Building," constituting 73% and 58.5%, respectively. Due to the BCD-oriented nature of FC-Siam-Diff and BITNet, these methods have difficulty identifying semantic changes in water, tree, and playground, as is evident from our visual analysis. In contrast, our proposed SCD-SAM exhibits exceptional performance, with its "From–To" matrix results closely mirroring the ground truth. This demonstrates the ability of our semantics-driven visual model to distinguish accurately and efficiently between different classes with high similarity. Due to fewer types of semantic changes and larger semantic change pixels, the results of various methods are more impressive in the Landsat-CD. The primary change types include "Water–Desert" and

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TABLE III
QUANTITATIVE RESULTS OF THE ABLATION ANALYSIS FROM THE SECOND-CD AND LANDSAT-CD. THE **RED** AND **BLUE** HIGHLIGHTS REPRESENT OPTIMAL AND SUBOPTIMAL VALUE. "W/O" REPRESENTS NETWORK WITHOUT SPECIFIC MODULE

Dataset	Methods	OA(%)	mIOU(%)	SeK(%)	F_{SCD}(%)
SECOND-CD	w/o SCD adaptor	85.41	68.77	16.50	56.76
	w/o MobileSAM encoder	85.74	69.08	15.81	55.51
	w/o CNN encoder	**87.52**	**72.44**	**21.49**	**61.23**
	w/o dual decoder	86.59	71.49	20.07	59.87
	SCD-SAM	**90.16**	**77.75**	**32.44**	**70.51**
Landsat-CD	w/o SCD adaptor	86.02	68.07	16.53	61.51
	w/o MobileSAM encoder	91.20	79.11	35.31	76.03
	w/o CNN encoder	**92.15**	80.05	36.77	77.14
	w/o dual decoder	91.78	**80.15**	**37.23**	**77.17**
	SCD-SAM	**93.30**	**83.04**	**44.05**	**81.16**

"Desert–Farmland," with change proportions of 91.3% and 79.4%, respectively. Compared with other methods, there are varying degrees of false negatives and false positives, while our approach achieves 92.7% and 79.5%, respectively, closely aligning with the ground truth.

E. Ablation Analysis

We performed ablation analysis on the SECOND-CD and Landsat-CD. Table III presents the quantitative results. It is evident that the performance on the four quantitative metrics across both datasets is not satisfactory when we remove the SCD adaptor and apply MobileSAM directly to SCD. This illustrates the difficulty it faces in segmenting remote sensing scenes with multiscale changes. This further emphasizes the effectiveness and necessity of our SCD adaptor. Therefore, by incorporating SCD adaptors into the network to fine-tune MobileSAM, we observe the significant performance improvements. The mIOU increases by 3.67% and 11.98% on the two datasets, indicating the feasibility of leveraging the powerful semantic-driven capabilities of VFMs in remote sensing scenes. In addition, when we remove the progressive feature aggregation dual decoder, the SeK metrics are only 20.07% and 37.23% on the two datasets, indicating that designing dedicated feature decoders for BCD and SS subtasks is crucial for the SCD task. Finally, our proposed SCD-SAM achieves mIOU values of 77.75% and 83.04% on two datasets, indicating that the dual encoder effectively combines the long-range modeling capability of MobileSAM encoder with the detail perception ability of the CNN encoder. This supplementary relationship facilitates the extraction of more comprehensive multiscale semantic change features.

V. CONCLUSION

We propose SCD-SAM for SCD by exploring the feasibility and effectiveness of semantic-driven VFMs. Specifically, the contextual semantic change-aware dual encoder consists of parallel MobileSAM and CNN encoders, extracting multiscale contextual semantic features from bitemporal images. In addition, we introduce a DFI module for injecting local features into the MobileSAM encoder, which serves as a bridge between the MobileSAM encoder and the CNN encoder. Progressive feature aggregation dual decoder is designed to aggregate multiscale features. Specifically, the binary decoder generates the binary change map as prior change information to guide SCD. Meanwhile, the Siamese semantic decoder progressively aggregates shallow and deep features across different levels, resulting in bitemporal SS maps. Based on the visual and quantitative analysis of the SECOND-CD and Landsat-CD, it has been demonstrated that the proposed SCD-SAM is capable of identifying various semantic changes accurately and achieving inspiring metric values. These results demonstrate the potential for transferring strong VFMs for remote sensing applications. Therefore, we posit that adaptation emerges as the most appropriate method for integrating SAM into SCD. As we advance general SCD in remote sensing images and develop new adaptation techniques, we aspire for this study to lay the foundation for forthcoming research endeavors.

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